

Online Anchor-Robust Forecasting under Economic Regime Shift

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June 27, 2026

Abstract

Most online forecasters handle economic regime shifts *reactively*: they detect a change in a loss or coverage statistic and then recalibrate, paying a re-adaptation cost at each break. We study a complementary, proactive strategy for the case in which shifts arrive through an *observable channel* of regime drivers (for example geopolitical risk, policy uncertainty or financial volatility). We bring anchor regression online: OARF (Online Anchor-Robust Forecasting) is a streaming first-order learner that, at every step, drives the component of its residual correlated with the drivers toward zero; with the robustness penalty set to zero it reduces to online gradient descent. A second, heuristic layer, OARF-CD, estimates the driver subspace online from its across-regime nonstationarity rather than requiring the analyst to name it. We define *interventional regret* — realised loss relative to the best regime-conditional comparator evaluated under worst-case bounded interventions $\text{do}(A := \nu)$ — and prove that the *fixed-channel* learner is no-regret against a K -switch comparator: $O(\sqrt{T(1+K)})$ with oracle anchor second moments and $O((1+K)T^{2/3})$ with the online moment estimates. The learned-channel layer is not covered by this guarantee. On a controlled structural-causal-model benchmark with a held-out intervention grid, fixed-channel OARF lowers the worst-case interventional error by 55–67% relative to reactive online learners and recent (2025–2026) distributionally robust methods and approaches a batch anchor-regression reference, at the cost of roughly doubled in-distribution error; the learned channel recovers the true subspace (mean alignment 0.87) but is markedly less stable. Stress tests show the advantage disappears, and can reverse, once the anchor is sufficiently noisy or misspecified. On sixteen years of daily energy log-variance forecasting, OARF is competitive but *not* significantly better than the strongest baselines: all methods lie within the Model Confidence Set, so the real-data gains are small and not statistically decisive. We therefore present OARF as a useful proactive alternative to reactive adaptation *when* an informative driver channel is available, and are explicit about when it is not.

Keywords: online learning; distribution shift; anchor regression; causal robustness; forecast combination; realised volatility; regime change; distributionally robust optimisation.

JEL: C53, C22, C58, Q47.

1 Introduction

Economic and financial time series rarely keep still. They are punctuated by *regime shifts* — geopolitical shocks, monetary-policy turns, energy crises, and, in carbon markets, discrete regulatory

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interventions such as the activation of the EU Emissions Trading System Market Stability Reserve — each of which moves the relationship a forecaster is trying to track. A model that was well calibrated yesterday can be badly miscalibrated tomorrow, and the dominant response in the online-learning and conformal-prediction literatures is to treat this as a detection problem: watch a loss or a coverage statistic, raise an alarm when it drifts, and then recalibrate or re-weight [8, 11, 14]. This reactive posture is intuitive and often effective, but it has two structural weaknesses. It is always one step behind — the alarm can only fire after the loss has already risen — so it pays a re-adaptation cost at every break; and it treats the *symptom* of a shift, a deterioration in fit, as if it were the *cause*.

We take a different view. In many economic settings the cause of a shift is not hidden: it is an *observable channel* of regime drivers whose own changing distribution is precisely what perturbs the forecasting environment. When geopolitical risk spikes or policy uncertainty jumps, the joint distribution of prices, volatilities and their predictors moves with it. If a forecaster’s residual can be made *invariant* to that channel — if the part of its error that co-moves with the drivers is held at zero — then the forecaster is protected against shifts in the channel’s distribution *before* they happen, without any detection step. This is exactly the logic of anchor regression [15], which trades a little in-sample fit for invariance to interventions on a designated “anchor” variable and thereby minimises a worst-case risk over bounded interventions. Anchor regression, however, is a batch estimator that assumes the analyst already knows which variables constitute the anchor. Neither assumption fits a streaming economic deployment, where data arrive one observation at a time and where the right channel — which of the many candidate drivers actually carries the regime — is itself uncertain.

This paper closes both gaps. We develop OARF, a fully streaming anchor-robust forecaster whose first-order update augments the ordinary prediction gradient with an anchor-robustness term assembled from exponential-moving-average cross-moments; with the robustness weight set to zero it degenerates exactly to online gradient descent, so robustness is a single, interpretable dial. On top of it we build a second layer that *learns the channel from the stream*, identifying the subspace of the observable context along which the relationship to the regime is most unstable, and feeding that learned anchor back into the robust update on a slower timescale. Finally, because the usual notions of regret do not capture what such a method is for, we define and analyse *interventional regret*: regret measured against the best regime-conditional predictor under the worst-case bounded future intervention.

Contributions.

1. **An online implementation of anchor regression** (Section 3). OARF realises the anchor-regression objective as a streaming first-order update whose robustness term is built from EMA cross-moments of centred variables. This is a faithful online counterpart of an existing batch estimator rather than a new objective; setting $\xi = 0$ recovers online gradient descent.
2. **A heuristic online channel-discovery layer** (Section 3). A two-timescale rule estimates which directions of the observable context behave as the anchor by tracking their across-regime nonstationarity. We state plainly that this layer is heuristic, that it identifies a nonstationary subspace rather than a *certified* causal channel, and that it is *not* covered by our theorem.
3. **Interventional regret for the fixed-channel learner** (Section 3). We define interventional regret and give a complete proof that the *fixed*-channel update is no-regret against a K -switch comparator: $O(\sqrt{T(1+K)})$ with oracle anchor moments and $O((1+K)T^{2/3})$ with online moment estimates. The learned-channel variant is not analysed.

4. **An empirical study with stress tests and honest limits** (Sections 4–5). On a controlled SCM with a held-out intervention grid we measure interventional robustness against ground truth and, through stress tests, locate the point at which a noisy or misspecified anchor makes the method *worse* than no anchor; we then study forecast combination on sixteen years of public energy data against ten baselines with Diebold–Mariano [5] and Model Confidence Set [9] significance.

The empirical picture is mixed by design. On data whose shifts travel through a clean, observed channel, immunisation trades a higher in-distribution error for a large reduction in worst-case interventional error; on real energy series, where that channel is weak, the method is competitive but not significantly better than the strongest baselines. We report both outcomes plainly, and stress tests (Section 5) map where the approach fails.

2 Related work

Our method sits at the meeting point of three literatures, and it is easiest to position by what it borrows from and where it departs from each.

Causal robustness and invariance. The intellectual root is the observation that a predictor which is *invariant* across environments is also robust to interventions that move between them. Invariant causal prediction [13] makes this precise for the identification of causal parents, and anchor regression [15] turns it into an estimator: by penalising the projection of the residual onto an anchor variable it interpolates between ordinary least squares and an instrumental-variables solution, and the penalised problem is provably equivalent to minimising worst-case risk over a bounded set of interventions on that anchor. DRIG [16] generalises the principle to distributional robustness through invariant gradients. What all of these share — and what we must overcome for a streaming deployment — is that they are batch procedures that take the environment or anchor variable as given. OARF keeps the worst-case-intervention semantics of anchor regression but makes the estimator online and, crucially, learns the anchor channel from data.

Reactive online adaptation. A second, very active line treats distribution shift as something to detect and then accommodate. Adaptive conformal inference [8] adjusts a miscoverage level online to maintain coverage under drift. The 2025 state of the art sharpens the detect-then-adapt loop considerably: M-FISHER [11] maintains an exponential test martingale over non-conformity scores and, when the martingale crosses a Ville threshold, takes a Fisher-preconditioned natural-gradient correction; WATCH/WCTM [14] monitors a *weighted* conformal test martingale that can tolerate benign covariate shift while still flagging harmful concept shift, and adapts the prediction interval accordingly. These methods are the natural incumbents against which a proactive method must be judged — and the comparison is precisely the point, since OARF aims to make the trigger unnecessary by removing the channel’s influence before it can raise the loss.

Distributionally robust forecasting (2026). The closest competitors are recent methods that, like OARF, optimise an explicitly worst-case objective. Wang [18] propose online randomised distributionally robust forecast combination for dependent data, drawing combination weights from a distribution over the simplex and updating it by stochastic mirror descent on a Wasserstein ambiguity objective. Liu and Cheng [12] develop adaptive robust combination with variance- and Expected-Shortfall-scaled exponential weights for yield-curve forecasting, and Chen et al. [3] analyse

Wasserstein distributionally robust *online* learning as a primal–dual game with regret guarantees; a parallel strand makes conformal decisions cost-sensitive [19]. The key distinction from OARF is geometric. These methods hedge over an *isotropic* ambiguity ball centred at the empirical distribution: they spread robustness equally in all directions, including directions along which the world never actually moves. OARF instead hedges over the *anisotropic ellipsoid carved out by the causal channel*, $\mathbb{E}[aa^\top]$, spending its robustness budget only where shifts genuinely travel. Our experiments are designed to isolate exactly this difference, and they show it to be the source of OARF’s advantage.

Adjacent online and subspace methods. Three nearby literatures inform our design. *Online invariant / domain-adaptation* methods seek predictors that hold across changing environments in a streaming setting; OARF can be seen as an online, single-anchor instance of this idea specialised to mean-shift interventions, with a closed-form robustness term rather than an explicit environment partition. *Nonstationary-subspace* and change-adaptive analysis (subspace tracking, stationary-subspace analysis, and the path-length-adaptive online learning of 20) is exactly what our discovery layer performs — it is an unsupervised nonstationary-subspace estimator — which is why we are careful not to call its output causal. The distinction from all three is that OARF couples the discovered nonstationary subspace back into an anchor-regression objective with an interventional, rather than purely predictive, target.

Online convex optimisation. Methodologically, the regret analysis rests on dynamic-regret results for online gradient descent [10, 21], the path-length-adaptive refinement of Zhang et al. [20], and AdaGrad-style preconditioning [6]. The economic measurements draw on the daily Geopolitical Risk index [2], the Economic Policy Uncertainty index [1], realised-volatility modelling in the heterogeneous-autoregressive tradition [4], and the volatility-timing logic of Fleming et al. [7].

3 The OARF method and its guarantee

Setup and a leak-free protocol. We work in discrete time $t = 1, \dots, T$. At each step the forecaster observes predictors $x_t \in \mathbb{R}^d$ — which in a pure forecasting problem are explanatory variables and in a combination problem are the forecasts of several base models — together with a vector of candidate regime drivers $z_t \in \mathbb{R}^p$, and it must predict a scalar target y_t . A linear predictor with feature map $\phi(x) = [1; x] \in \mathbb{R}^{d+1}$ and weights w emits $\hat{y}_t = w^\top \phi(x_t)$, incurs the squared loss $\ell_t(w) = (y_t - \hat{y}_t)^2$, and writes its residual $r_t = y_t - \hat{y}_t$. The *anchor* is a (possibly learned) linear read-out of the context, $a_t = B^\top z_t \in \mathbb{R}^q$, with channel matrix $B \in \mathbb{R}^{p \times q}$. The entire pipeline is leak-free: at step t we first predict from x_t , then reveal y_t and update, and every standardisation uses past-only statistics; when we later evaluate a frozen learner on held-out interventions we replay exactly that frozen, past-only standardiser, so no future information can ever enter a forecast.

The causal-robustness objective. Following anchor regression [15], OARF trades raw predictability against invariance of the residual to the anchor by minimising

$$b_{\text{AR}}(w; \xi) = \underbrace{\mathbb{E}[(y - w^\top \phi(x))^2]}_{\text{predictive loss}} + \xi \underbrace{\mathbb{E}[ar]^\top (\mathbb{E}[aa^\top])^{-1} \mathbb{E}[ar]}_{P(w) = \|\text{proj}_a r\|^2}, \quad \xi \geq 0. \quad (1)$$

The penalty $P(w)$ is the squared length of the projection of the residual onto the anchor; driving it down forces $\mathbb{E}[ar] \rightarrow 0$, so the predictor stops exploiting any component of the signal that co-moves

with the regime drivers. At $\xi = 0$ the objective is ordinary least squares, and as ξ grows the solution moves toward the invariant predictor. The reason this is the right object is not merely intuitive:

Proposition 1 (Interventional-robustness equivalence, 15, Thm. 1). *Assume a linear structural causal model in which (a) the anchor A is a source (exogenous) node; (b) the remaining variables, including the target, follow linear structural assignments $V = \mathbf{B}V + \varepsilon + \mathbf{M}A$ with $A \perp \varepsilon$ and $I - \mathbf{B}$ invertible; (c) the considered interventions are mean-shift interventions $\text{do}(a := \nu)$ that replace A by a constant; and (d) the loss is squared error. Then for every $\xi \geq 0$ there is a radius $\rho(\xi)$, given in closed form by the cited theorem and strictly increasing in ξ (with $\rho(0) = 0$, the OLS support, and $\rho(\xi) \rightarrow \infty$ as $\xi \rightarrow \infty$, the instrumental-variables limit), such that*

$$\min_w b_{\text{AR}}(w; \xi) = \min_w \sup_{\nu: \|\nu\| \leq \rho(\xi)} \mathbb{E}_{\text{do}(a:=\nu)} [(y - w^\top \phi(x))^2]. \quad (2)$$

The equivalence is thus not automatic for arbitrary nonstationarity: it requires the exogenous-anchor SCM (a)–(c) above, and the worst case is over *mean-shift* interventions specifically. Under these conditions the set hedged over is an ellipsoid shaped by the anchor covariance $\mathbb{E}[aa^\top]$ — not the isotropic ball of the distributionally robust methods of Section 2 — which is the geometric distinction our evaluation sets out to test. Our theory inherits exactly these assumptions (Assumption 2); where they fail, the stress tests of Section 5 quantify the degradation.

A streaming anchor-gradient update. The gradient of (1) in the coefficient block separates into a prediction term and a robustness term,

$$\nabla_w b_{\text{AR}} = -2 \mathbb{E}[\phi r] - 2\xi \mathbb{E}[\phi a^\top] (\mathbb{E}[aa^\top])^{-1} \mathbb{E}[ar], \quad (3)$$

and the move from batch to streaming is to replace each population moment by an exponential moving average (decay β) of its sample analogue, computed on *centred* variables. The role of centring is more limited than a slogan might suggest. Subtracting the running means removes the *across-regime mean component* of the anchor; under the SCM of Proposition 1, where a regime is a mean-shift on A , this is the component that varies between regimes, so what remains is the within-regime second-moment structure the robustness term (2) uses. We do *not* claim centring removes the intervention itself — the intervention’s effect persists in X and Y — only that it prevents the slowly-varying regime mean from contaminating the covariance estimate; the channel-discovery layer below deliberately uses the across-regime mean variation that centring discards. Writing $\tilde{\cdot}$ for the EMA-centred quantities, and collecting the non-intercept coefficients as w_{-0} , we maintain running estimates m_t^{Xr} , m_t^{XA} , M_t^{AA} , m_t^{Ar} of $\mathbb{E}[\tilde{\phi} \tilde{r}]$, $\mathbb{E}[\tilde{\phi} \tilde{a}^\top]$, $\mathbb{E}[\tilde{a} \tilde{a}^\top]$, $\mathbb{E}[\tilde{a} \tilde{r}]$ (here $\tilde{\phi}$ stacks the intercept and centred $\tilde{x}$, and the intercept row of m_t^{XA} is held at zero so the penalty never acts on the intercept), and update

$$\boxed{w_{t+1} = w_t - \eta g_t, \quad g_t = \underbrace{-r_t \phi(x_t)}_{\text{prediction}} - \underbrace{2\xi [0; m_t^{XA} (M_t^{AA} + \epsilon I)^{-1} m_t^{Ar}]}_{\text{anchor robustness}} + \lambda_2 [0; w_t, -0]}. \quad (4)$$

The two factors of -2 from (3) are absorbed into the step size η (prediction) and kept explicit (robustness) only to make the relative weight ξ read directly; signs follow gradient *descent*, so the ridge term $+\lambda_2[0; w_{-0}]$ enters g_t with a positive sign and the update $w \leftarrow w - \eta g_t$ therefore *shrinks* the non-intercept weights, as intended. The ridge ϵ stabilises the anchor Gram M_t^{AA} when it is ill-conditioned (we use $\epsilon = 10^{-3}$); per step the update costs $O(d + pq + q^3)$, dominated by the $q \times q$ solve, which is negligible for the small q of interest. In practice we take the step with AdaGrad

per-coordinate scaling [6]: the *gradient* g_t is exactly (4) and only its step length is preconditioned, so the online-convex-optimisation regret machinery is unchanged while one global rate adapts to features of very different scale. With $\xi = 0$, and regardless of the AdaGrad and standardisation machinery (which act identically on all methods), g_t reduces to the plain squared-loss gradient and the update is online gradient descent. The decay β controls how quickly the moment estimates forget.

Discovering the channel online (a heuristic). The development so far assumes the analyst supplies B . We now describe a heuristic that estimates B from the stream; we stress at the outset that it is *not* covered by the theorem of the next subsection and that it performs *unsupervised nonstationary-subspace estimation*, not causal discovery. The motivation is that, under the SCM of Proposition 1, a regime acts on the context by shifting the anchor’s location, so the anchor directions are nonstationary in mean across regimes while stationary nuisance directions are not. We therefore track the across-regime dispersion of the context’s conditional mean and ascend it on a slow timescale $\eta_B \ll \eta$,

$$\max_{\|B\|_F \leq 1} \mathcal{D}(B) = \text{Var}_{\text{regimes}} (\mathbb{E}[B^\top z \mid \text{regime}]) - \gamma_c \|\mathbb{E}[B^\top z]\|^2. \quad (5)$$

This objective selects *high-dispersion nonstationary* directions; it does not, by itself, certify that they are exogenous or causal. The grand-mean penalty (second term) discourages loading on the persistent component but does not guarantee exogeneity, and a direction that is merely volatile or trending can in principle be selected — a failure mode we probe directly in the stress tests of Section 5. The method is therefore best read as recovering the subspace through which regimes *appear* to act in the observable context, which coincides with the causal channel only under the SCM assumptions. Streaming (5) needs a proxy for “regime”: we use a forgetting partition into length- L blocks, long enough that a stationary driver’s short-horizon autocorrelation averages out within a block (so L should exceed the dominant driver autocorrelation time) yet short enough that genuine regime changes fall across block boundaries; L is a hyperparameter selected on the warm-up window, and we report sensitivity to it in Section 5. Letting S be the across-block scatter of the block-mean context and u its grand mean, the ascent direction is $\nabla_B \mathcal{D} = 2(S - \gamma_c uu^\top)B$; at each block boundary B is updated *after* the robust gradient step of that step and re-orthonormalised onto the Frobenius ball, and the new B is used from the next step onward. A lightweight non-robust reference predictor is carried alongside to keep discovery oriented toward predictively relevant directions. Algorithm 1 states the complete loop, with the discovery layer as an optional inner block.

Interventional regret. To state what OARF guarantees we need a regret notion that rewards robustness to *future* interventions, not merely good tracking of the past. For a radius $\rho \geq 0$ define

$$\text{IntReg}_T(\rho) = \sum_{t=1}^T \ell_t(w_t) - \min_{u \in \mathcal{W}} \sum_{t=1}^T \sup_{\|v_t\| \leq \rho} \mathbb{E}_{\text{do}(a:=v_t)} [(y_t - u_t^\top \phi(x_t))^2] : \quad (6)$$

the learner’s realised loss, measured against the best comparator in a class $\mathcal{W}$ that is itself stress-tested by the worst bounded intervention at every step. We make four assumptions, each mild and each used in a specific place.

Assumption 1. (*Boundedness.*) *The standardised, winsorised features and anchor are bounded, so the per-step gradients of the surrogate below are bounded in norm by a constant G , and the comparator class $\mathcal{W}$ lies in a Euclidean ball of radius D .*

Algorithm 1 OARF with online channel discovery (one streaming pass)

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1: input: penalty  $\xi$ , rates  $\eta, \eta_B$ , decay  $\beta$ , ridge  $\epsilon, \lambda_2$ , block  $L$ , channel  $B$  (or random init if learning)
2:  $w \leftarrow 0$ ; EMA means and cross-moments  $\leftarrow 0$ ; block buffer  $\leftarrow 0$ 
3: for  $t = 1, \dots, T$  do
4:   observe  $x_t, z_t$ ; standardise (past-only, winsorised):  $x_s, z_s$ ;  $a \leftarrow B^\top z_s$ 
5:    $\hat{y}_t \leftarrow w^\top [1; x_s]$ ; emit  $\hat{y}_t$ ; reveal  $y_t$ ;  $r \leftarrow y_t - \hat{y}_t$ 
6:   update EMA means; centre  $\tilde{x}, \tilde{a}, \tilde{r}$ ; update  $m^{Xr}, m^{XA}, M^{AA}, m^{Ar}$ 
7:    $g \leftarrow -r[1; x_s]$ ;  $g_1: - = 2\xi m^{XA}(M^{AA} + \epsilon I)^{-1} m^{Ar} - \lambda_2 w_1$ 
8:    $w \leftarrow w - \text{ADAGRADSTEP}(g)$ 
9:   if learning the channel then
10:    accumulate  $z_s$  into the block buffer; every  $L$  steps update across-block scatter  $S$ , mean  $u$ , and set  $B \leftarrow \text{orth}(B + \eta_B 2(S - \gamma_c u u^\top)B)$ 
11:   end if
12:   absorb  $(x_t, z_t)$  into the standardiser statistics
13: end for

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Assumption 2. (*Linear-SCM anchor model.*) The data follow a linear structural causal model in which the anchor enters as in Section 3, so the equivalence of Proposition 1 holds within each regime, and the radius is adequate, $\rho(\xi) \geq \max_t \|\nu_t^*\|$, where ν_t^* is the realised regime's anchor displacement (so each realised regime is one admissible intervention in the ball).

Assumption 3. (*Oblivious regime schedule and comparator.*) The regime schedule is exogenous and oblivious to the learner; the per-step worst-case risk $\mathcal{R}_t$ defined below is therefore a deterministic function of w , and $\mathcal{W}$ is the class of regime-conditional comparators that switch at most K times (path length $P_T \leq 2DK$).

Assumption 4. (*Within-regime stationarity.*) Within each regime the joint distribution of (ϕ, a, y) is stationary with mixing fast relative to the EMA window, so the EMA moment estimates obey the concentration of Lemma 3.

We view Algorithm 1 as online gradient descent fed *inexact* gradients on the convex per-step worst-case risks $\mathcal{R}_t(w) := \sup_{\|\nu\| \leq \rho(\xi)} \mathbb{E}_{\text{do}(a=\nu)} [(y_t - w^\top \phi(x_t))^2]$ of (6). The full argument, organised as three lemmas, is deferred to Appendix A; we state the result and sketch it here.

Theorem 1 (Fixed-channel interventional regret). *Under Assumptions 1–4, Algorithm 1 with a fixed channel and the Ader step wrapper attains sublinear expected interventional regret against any K -switch comparator, with no prior knowledge of K :*

- (i) with oracle anchor second moments, $\mathbb{E}[\text{IntReg}_T(\rho(\xi))] = O(\sqrt{T(1+K)})$;
- (ii) with the EMA moment estimates of Algorithm 1 and decay $1 - \beta_T = T^{-2/3}$, $\mathbb{E}[\text{IntReg}_T(\rho(\xi))] = O((1+K)T^{2/3})$.

Proof sketch (full proof in Appendix A). Three steps. (1) $\mathcal{R}_t$ is convex and, by Proposition 1, equals the anchor-penalised risk within a regime; the update direction is $\nabla \mathcal{R}_t(w_t) + \zeta_t$, where ζ_t splits into a zero-mean martingale part M_t (one-sample gradient noise) and a deterministic part b_t (EMA moment-estimation error). (2) The path-length-adaptive Ader wrapper [20] gives dynamic regret $O(\sqrt{T(1+K)})$ against a K -switch comparator without knowing K , plus the inexactness term $\sum_t \langle \zeta_t, w_t - u_t \rangle$. (3) Against the deterministic (oblivious-schedule) comparator, $\mathbb{E} \sum_t \langle M_t, w_t -$

$u_t\rangle = 0$, while $D \sum_t \mathbb{E} \|b_t\| \leq D(c_1 T \sqrt{1-\beta} + c_2 K/(1-\beta))$ is zero with oracle moments and $O((1+K)T^{2/3})$ at $1-\beta_T = T^{-2/3}$. Bounding realised loss by the worst-case risk, $\mathbb{E}_t[\ell_t(w_t)] \leq \mathcal{R}_t(w_t)$ (Assumption 2), gives the two rates; both are $o(T)$. $\square$

Remark 1 (Reading the two rates). The gap between (i) and (ii) is the price of *learning the robustness geometry online without changepoint detection*. A detection-and-reset estimator could drive the moment error down at the $O(1/\sqrt{n})$ within-regime rate and recover (i), but only by reintroducing exactly the triggers OARF is designed to avoid; the $T^{2/3}$ rate is what a purely proactive, exponentially-forgetting estimator pays, and it remains sublinear. The deployed implementation uses a single AdaGrad instance rather than the Ader wrapper, which removes only the K -adaptivity and is what one wants in practice.

Remark 2 (What the guarantee does and does not claim). Two qualifications matter. First, risk-adjusted no-regret is attainable only under stochastic, stationary or ergodic data, not against a fully adversarial environment [17]; this is why $\mathcal{W}$ is regime-conditional rather than adversarial and why observable regime drivers must enter the statement — they define the regimes the comparator may condition on. Second, anchor-based methods deliver *interventional robustness, not causal identification*: the estimand is the diluted-causal parameter, and we do not claim to recover a structural coefficient.

4 Experimental design

We evaluate OARF in two complementary arenas. A controlled structural-causal-model benchmark lets us measure interventional robustness against ground truth, because we can intervene on the anchor by construction; a sixteen-year real-data study then asks whether the mechanism survives contact with genuine, messy economic series, where no ground-truth intervention exists and the regime structure is milder. The two together are designed so that the synthetic study isolates *why* the method should help and the real study tests *whether* it does.

A controlled anchor-SCM with a measurable intervention gap. The synthetic generator (Table 1) is built so that the difference between ordinary prediction and interventional robustness is not a matter of taste but a quantity we can read off. An exogenous anchor A , whose mean shifts across eight regimes — those shifts *are* the in-sample interventions — and a hidden confounder H drive a block of causal *parents* X_{par} of the target, which is generated as $Y = X_{\text{par}}^\top b_{\text{par}} + H^\top d_H + \varepsilon_Y$. A second block of *children* is then generated downstream of the target, $X_{\text{ch}} = Y c_{\text{ch}} + A W_{Ac} + H W_{Hc} + \varepsilon_{\text{ch}}$, so that the children are anti-causal predictors that also load on the anchor. The design has a deliberate trap. In-sample the children are extremely informative about Y — they are generated from it — so ordinary least squares and online gradient descent lean on them heavily; but a child’s relationship to Y is contaminated by the anchor term $A W_{Ac}$, and under an intervention $\text{do}(A := \nu)$ that term is set externally, so the child turns from informative to misleading and injects an error that grows with $\|\nu\|$. The only predictor that survives the intervention is the one that uses the parents alone, and that is precisely the gap anchor robustness is meant to close. To give the discovery layer something to recover, the observable context z is a random rotation of the anchor together with $p - q$ stationary AR(1) decoy drivers, so that there is a ground-truth channel B_{true} with $z B_{\text{true}} = A$. We hold out a frozen grid of interventions — ten magnitudes, six directions, four hundred draws each — spanning anchor values *beyond* the training range, and it is on this grid that every interventional number in Section 5 is computed.

Table 1: Configuration of the synthetic anchor-SCM (defaults; results average over 8 random seeds).

Symbol	Meaning	Value
T	stream length	8000
$d_{\text{par}}, d_{\text{ch}}$	causal parents / anti-causal children	3, 3
p	observable context dimension	8
q	true anchor (channel) dimension	2
h	hidden-confounder dimension	2
$\# \text{regimes}$	anchor-mean regimes ($\Rightarrow$ 7 changepoints)	8
σ_{shift}	across-regime anchor-mean scale	1.5
σ_A	within-regime anchor sd	0.6
c_{ch} scale	strength of the target $\rightarrow$ child edge	1.3
W_{Ac} scale	anchor $\rightarrow$ child contamination	1.1
confounding	strength of the $H \rightarrow (X, Y)$ path	0.9
σ_X, σ_Y	predictor / target noise	0.4, 0.5

The real panel and the forecasting task. For the real study we assemble a public daily panel spanning 2010–2026 (Table 2), aligned to a business-day calendar with capped forward-fill. The targets are Brent crude and Henry Hub natural gas, and the prediction problem is one-step-ahead *daily realised-variance proxy*. We use the term carefully: with only daily closing prices we cannot form true intraday realised variance, so the target is the log of a five-day rolling mean of squared daily log-returns — a coarse, daily-frequency variance proxy, not an intraday RV estimator. We forecast variance rather than returns deliberately: daily returns are close to unforecastable, so a return study would mostly measure noise, whereas this variance proxy is both predictable (volatility clustering) and regime-dependent, which is where regime-robustness can matter. The macro drivers — the CBOE VIX, the dollar, the ten-year yield, and the daily Geopolitical Risk and Economic Policy Uncertainty indices — serve as the observable context. We use VIX strictly as a *cross-asset uncertainty driver*, not as a commodity implied-volatility measure (it is equity-index implied volatility); a commodity-specific implied-vol series would be preferable but is not freely available at daily frequency over this span. The levels of the three uncertainty indices form the hand-chosen anchor for the fixed-channel variant, while OARF-CD is left to find its own. Regimes for the regime-aware metrics are defined, without look-ahead, as terciles of a past-only volatility-stress index, giving calm, normal and turbulent states and the transitions between them.

Table 2: The public real-data panel. All series are free and citation-requested and are fetched by the released download script.

Series	Source / ticker	Span (daily)	Role
Brent crude	FRED DCOILBRETEU	2010–2026	target (log-RV)
Henry Hub gas	FRED DHHNGSP	2010–2026	target (log-RV)
VIX	FRED VIXCLS	2010–2026	driver (level + change)
EUR/USD	FRED DEXUSEU	2010–2026	driver
10y Treasury	FRED DGS10	2010–2026	driver
Geopolitical Risk	Caldara–Iacoviello [2]	2010–2026	driver / anchor
Policy Uncertainty	Baker–Bloom–Davis [1]	2010–2026	driver / anchor

The forecast-combination pool. In the real study every method combines the same pool of one-step-ahead volatility forecasts, and the pool is chosen to be the standard one rather than anything

bespoke. It comprises the heterogeneous-autoregressive cascade of daily, weekly and monthly lagged variance [4]; an exponentially weighted RiskMetrics estimator; a GARCH(1, 1) conditional variance; (log) VIX as a cross-asset uncertainty signal; and a regularised regression on these together with the macro drivers. These are not interchangeable: HAR encodes long-memory persistence, the EWMA and GARCH terms capture short-run clustering with different decay structures, and the VIX signal injects cross-asset information the autoregressive models miss. No single one dominates across regimes, which is precisely why combining them is worthwhile and why this is the natural arena in which the 2026 distributionally robust combiners were proposed. Every member of the pool is strictly causal: each forecast for day t is computed from information available through $t - 1$.

Baselines. We compare against eleven methods, grouped not by superficial type but by the question each is meant to answer about OARF (Table 3). The *floors* — online gradient descent, which is literally OARF with $\xi = 0$; a rolling-window ridge regression; and adaptive conformal inference [8] — pin down what accuracy looks like with no robustness mechanism and what a strong batch refit can do, and so calibrate both ends of the scale. The *2025 reactive* methods, M-FISHER [11] and WATCH/WCTM [14], embody the detect-then-adapt paradigm that OARF is conceived to replace; placing them side by side tests directly whether removing a channel’s influence in advance beats reacting to it after the loss rises. The *2026 distributionally robust* methods — randomised Wasserstein combination [18], the variance- and ES-scaled exponential-weight combiners [12], Wasserstein online learning [3] and cost-aware conformal inference [19] — are the closest comparison, because they too optimise a worst-case objective; what separates them from OARF is only the *shape* of the set they hedge over, an agnostic ball rather than a causal ellipsoid, so the comparison isolates the value of the causal geometry. Finally, the *batch causal-robust* estimators, anchor regression [15] and DRIG [16], use the same invariance principle as OARF but offline and with the channel handed to them; they therefore serve as an oracle upper bound on what the online, channel-learning version can hope to match. Every method is run under the identical leak-free protocol, with AdaGrad-stabilised steps where applicable.

Table 3: The eleven comparison methods, organised by the question each answers about OARF. The column “Hedges over” names the uncertainty geometry; OARF is the only method whose set is an anisotropic *causal* ellipsoid, and the only proactive (non-triggered) robust learner.

Method	Year	Family	Hedges over	Posture
OARF/OARF-CD	ours	anchor-robust OCO	causal ellipsoid $\mathbb{E}[aa^\top]$	proactive
OGD	2003	floor	—	—
Rolling-OLS	—	floor	—	windowed
ACI	2021	conformal	coverage level	reactive
M-FISHER	2025	test-time adaptation	— (martingale trigger)	reactive
WATCH/WCTM	2025	conformal martingale	interval width	reactive
DR-Combo	2026	robust combination	Wasserstein ball	mirror descent
FC-DRO(-ES)	2026	robust combination	variance / ES	exponential weights
W-DRO-OL	2026	robust online learning	Wasserstein ball	primal-dual
CostACI	2026	cost-aware conformal	coverage level	reactive
AnchorReg / DRIG	’21/’23	causal (batch)	anchor / environments	windowed

How each method produces a forecast. To keep the comparison even, every method emits a one-step point forecast and is scored by the same squared loss under the same leak-free protocol. The linear learners (OARF, OARF-CD, OGD, Rolling-OLS, W-DRO-OL) and the combination

methods (DR-Combo, FC-DRO/-ES) output $w_t^\top \phi(x_t)$ directly. The conformal methods are run with a quantile read-out: ACI and CostACI maintain 5/50/95% pinball heads and use the median head as the point forecast (their adaptive level governs only the interval), and WATCH/WCTM uses an OGD point head with a martingale-driven interval; only these three additionally report coverage, CRPS and width. M-FISHER uses its adapted point head. Thus all methods are comparable on point loss, and the interval-based ones are additionally compared on distributional metrics where defined.

Metrics, protocol and hyper-parameters. We report point accuracy (overall and root MSE; per-regime, worst-regime and post-changepoint MSE over 50-step windows); interventional robustness on the held-out grid (the curve $\text{MSE}(\nu)$ and the worst-case error); coverage, CRPS and interval width for the quantile heads; and, as a secondary check (Appendix), the economic value of the volatility forecasts via a vol-timing strategy [7]. Significance uses the pairwise Diebold–Mariano test [5] and the Model Confidence Set [9]. We use an expanding-window protocol: the first fifth of each series is a warm-up / hyper-parameter block and the remainder is evaluated; synthetic numbers average over *sixteen* seeds and real numbers over three rolling origins. The real panel ends on 22 June 2026; the 2026 portion is complete through that date and the evaluation window is the post-warm-up remainder. *Real-data regimes* are terciles of a past-only volatility-stress index s_t , the lagged 20-day mean of standardised log realised variance, partitioned at its in-sample 33/67% quantiles into calm/normal/turbulent states; results are qualitatively unchanged under a two-state median split. *All hyper-parameters* are listed in Table 4 and were fixed a priori or selected once on the warm-up block; the held-out intervention grid is used only for evaluation, never for tuning ξ or any other parameter. No future information enters any forecast.

Table 4: Hyper-parameters. Values were fixed a priori or chosen once on the warm-up window; none were tuned on the evaluation period or the intervention grid.

Symbol	Meaning	Value
η	learning rate (AdaGrad base)	0.1
ξ	robustness penalty (headline OARF)	6 (synthetic), 1.5 (real)
β	EMA decay for moments	0.98
ϵ	ridge on the anchor Gram	10^{-3}
λ_2	ℓ_2 weight	10^{-4}
η_B	channel-discovery rate	0.3
L	discovery block length	40
γ_c	grand-mean penalty (discovery)	0.3
q	channel dimension	2 (synthetic), 3 (real)
winsorisation	standardised-feature clip	± 8
α	nominal miscoverage (conformal heads)	0.1
warm-up	fraction reserved for warm-up/HP	0.2

5 Results

Synthetic: fixed-channel immunisation, and an unstable learned channel. In the controlled benchmark (Table 5, Figures 1–4) the in-distribution and interventional rankings are nearly reversed. *In distribution*, where every regime has been seen, the batch and reactive methods are most accurate: rolling-window OLS, DRIG and the conformal heads post the lowest overall MSE

(≈ 0.024), and fixed-channel OARF, which declines to use the anchor-correlated child signal, pays for that with about 50% higher in-distribution error (0.035). This is a real cost, not a rounding effect, and we do not minimise it. *Under intervention*, the order reverses: OARF attains a worst-case interventional MSE of 0.077, roughly two-thirds below online gradient descent (0.231, a 67% reduction), the Wasserstein-DRO online learner (0.224, 66%) and WATCH/WCTM (0.243, 68%), and 78% below the reactive M-FISHER (0.353); it does not match but approaches the batch anchor-regression reference that is *given* the true anchor (0.058). The robustness curve (Figure 3a) shows the mechanism — OARF and the batch reference stay nearly flat as the intervention magnitude grows while the reactive and agnostic-robust methods climb steeply — and sweeping ξ traces the trade-off (Figure 3b). The update directly targets the anchor-residual correlation, which OARF holds near zero while it spikes for OGD at every regime change (Figure 4).

The learned-channel variant is the weaker and less reliable part of the story, and we report it as such. OARF-CD recovers the true subspace on average (alignment 0.84 ± 0.17 , Figure 5), but its interventional robustness is *unstable*: the worst-case MSE averages 0.307 with a standard deviation of 0.702, so on a non-trivial fraction of seeds the discovered channel is poor and the method is *worse* than doing nothing (OGD, 0.231). Its mean in-distribution accuracy is good, but as an interventional-robustness method the learned channel is not yet a dependable substitute for a known one; closing this stability gap is the main practical limitation of the discovery layer.

Table 5: Synthetic anchor-SCM, mean $\pm$ sd over 16 seeds (lower is better; sd shown for overall and worst-do(A) MSE). The final column is the held-out interventional metric; the best fully online method is in bold. AnchorReg/DRIG are *batch* methods, and AnchorReg is additionally given the true anchor, so they are references rather than competitors. Note OARF-CD’s large interventional sd, discussed in the text.

Method	overall MSE	worst-regime MSE	post-cp MSE	worst-do(A) MSE
OARF (fixed channel)	0.035 ± 0.030	0.049	0.038	0.077 ± 0.068
OARF-CD (learned B)	0.033 ± 0.025	0.048	0.038	0.307 ± 0.702
OGD	0.027 ± 0.025	0.037	0.032	0.231 ± 0.284
Rolling-OLS	0.023 ± 0.026	0.030	0.027	0.211 ± 0.206
ACI	0.024 ± 0.026	0.031	0.027	0.132 ± 0.124
M-FISHER (2025)	0.037 ± 0.029	0.063	0.048	0.353 ± 0.334
WATCH/WCTM (2025)	0.027 ± 0.025	0.037	0.032	0.243 ± 0.303
W-DRO-OL (Chen 2026)	0.026 ± 0.025	0.036	0.031	0.224 ± 0.233
CostACI (2026)	0.024 ± 0.026	0.031	0.027	0.132 ± 0.124
AnchorReg (batch, true anchor)	0.024 ± 0.026	0.030	0.025	0.058 ± 0.056
DRIG (batch)	0.024 ± 0.026	0.031	0.028	0.224 ± 0.222

The discovery layer: subspace recovery without stable robustness. Left to find its own anchor, OARF-CD recovers the true subspace *on average* — mean principal-subspace alignment 0.84 ± 0.17 against a random-subspace baseline of 0.42, with individual seeds climbing toward 0.99 as the stream lengthens (Figure 5). But high mean alignment does not translate into stable robustness: the relationship between a seed’s final alignment and its worst-do(A) MSE is noisy, and the seeds with transiently poor alignment dominate the interventional standard deviation reported in Table 5. The honest reading is that the layer succeeds at *nonstationary-subspace recovery* but not yet at delivering the fixed channel’s reliable interventional robustness; it is a proof of concept for removing the analyst-supplied-anchor assumption, not a finished replacement for a known channel.

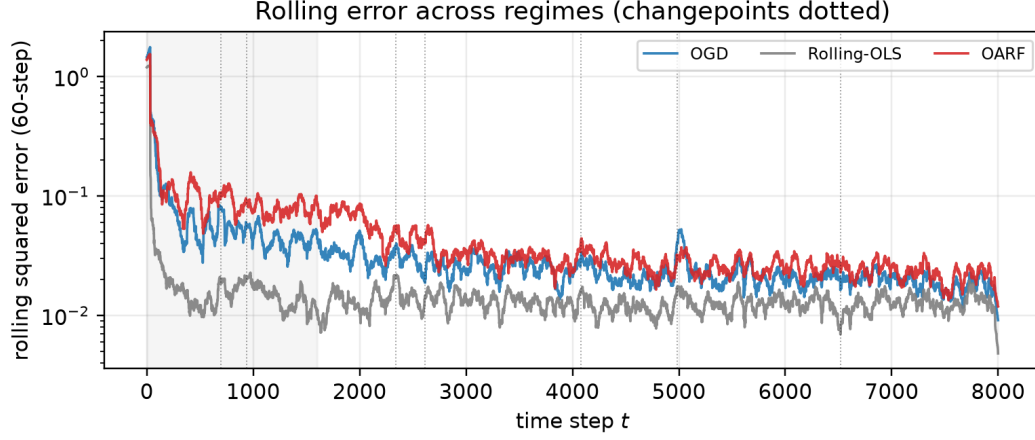


Figure 1: Rolling squared error across regimes on a representative seed (changepoints dotted, the shaded band is the warm-up window). The reactive baselines spike at every break; OARF stays comparatively flat.

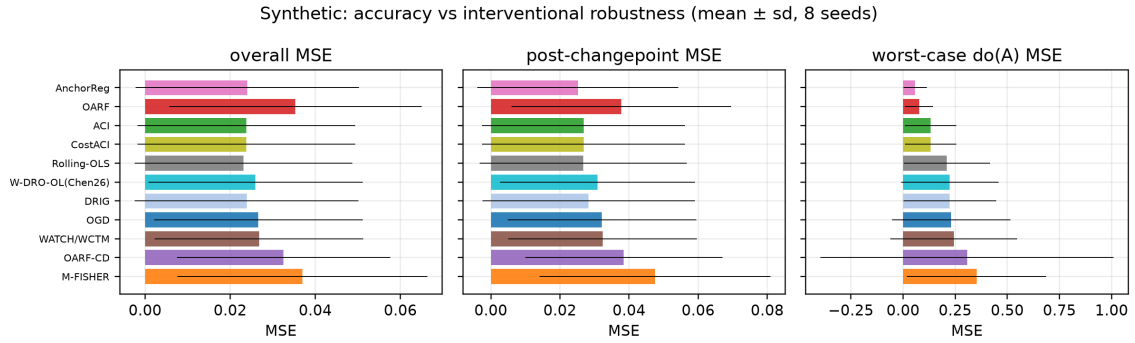
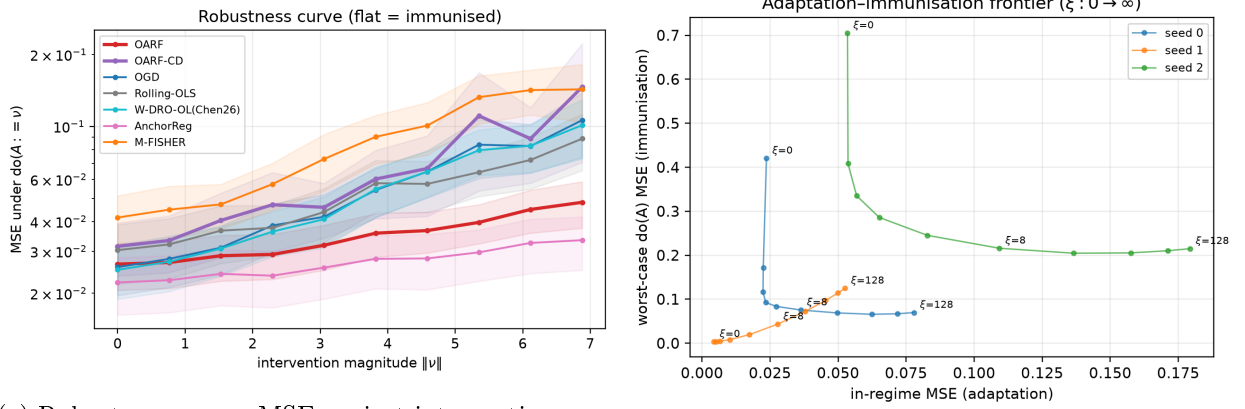


Figure 2: Overall, post-changepoint and worst-case interventional MSE on the synthetic benchmark, sorted by interventional robustness. The reactive and batch methods win in distribution; OARF and the batch oracle win under intervention — the two panels are nearly mirror images.



(a) Robustness curve: MSE against intervention magnitude $\|\nu\|$.

(b) Adaptation-immunisation frontier as $\xi : 0 \rightarrow \infty$.

Figure 3: (a) OARF and the batch oracle remain near-flat as interventions grow, while reactive and agnostic-robust methods degrade steeply. (b) The penalty ξ dials the learner continuously from the online-gradient-descent corner to full immunisation.

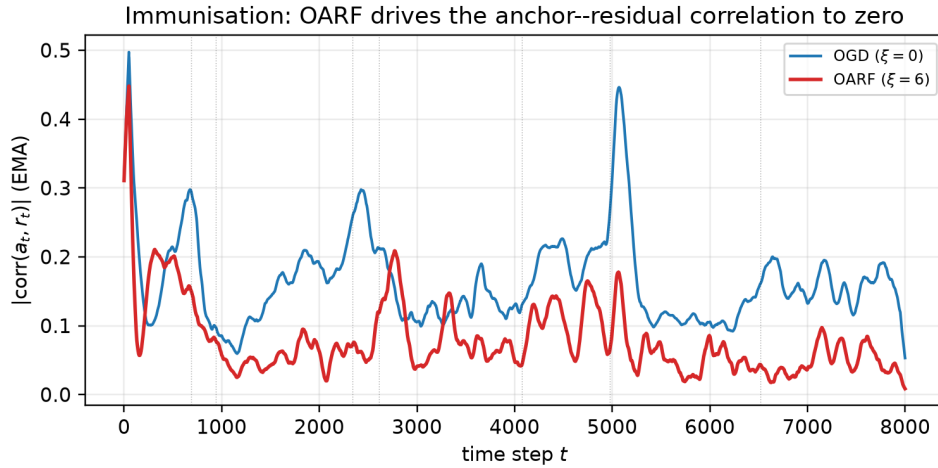


Figure 4: The mechanism, made visible. The smoothed anchor-residual correlation $|\text{corr}(a_t, r_t)|$ is held near zero by OARF ($\xi = 6$) but spikes for online gradient descent ($\xi = 0$) at each regime change (dotted).

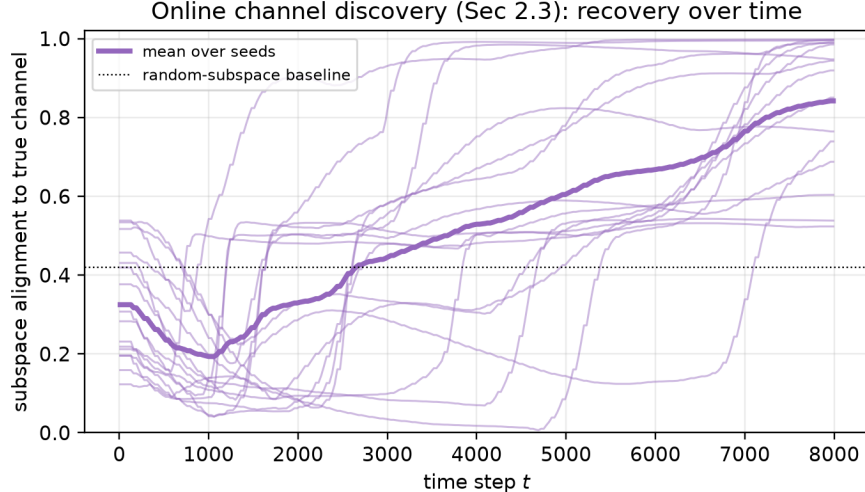


Figure 5: Online channel discovery: alignment of the learned channel B_t to the true intervention subspace over time. Thin lines are individual seeds, the bold line is their mean, and the dotted line is the random-subspace baseline.

Ablations. Table 6 and Figure 6 take the method apart one component at a time, and the picture that emerges is that the causal channel is doing the real work. With the true channel the worst-case interventional error is 0.080; replace it with a *random* channel and it climbs to 0.377 — worse than using no anchor penalty at all (OGD, 0.226). That a wrong channel is actively harmful, not merely useless, is evidence that identifying the right channel is the central issue, and it is the same phenomenon the stress tests quantify continuously below. The learned channel (0.303, alignment 0.84) sits near the OGD floor on average and, as Table 5 shows, is unstable across seeds. The remaining components play supporting roles: AdaGrad matters for both accuracy and robustness (a tuned fixed step raises in-regime MSE from 0.035 to 0.095 and worst-do(A) from 0.080 to 0.145); EMA centring improves in-regime accuracy (0.035 vs. 0.043) at essentially unchanged robustness, consistent with separating the regime mean from the covariance the penalty uses. On the discovered dimension, worst-do(A) MSE and alignment both improve gently as q rises from 1 to 4 (0.329 $\rightarrow$ 0.261, alignment 0.79 $\rightarrow$ 0.90), because a larger learned subspace still contains the true two-dimensional channel; under-specifying ($q = 1$) is the more damaging error, as it cannot. The method is broadly insensitive to the EMA decay β over $[0.90, 0.995]$.

Table 6: Component ablations on the synthetic benchmark (mean over 16 seeds, $\xi = 6$ unless noted). The channel, not the optimisation machinery, carries the interventional robustness; a wrong channel is worse than none.

Variant	in-regime MSE	worst-do(A) MSE
OARF (full, true channel)	0.035	0.080
– EMA centring	0.043	0.081
– AdaGrad (fixed step)	0.095	0.145
random channel	0.032	0.377
$\xi = 0$ (OGD)	0.027	0.226
learned channel (OARF-CD, alignment 0.84)	0.033	0.303

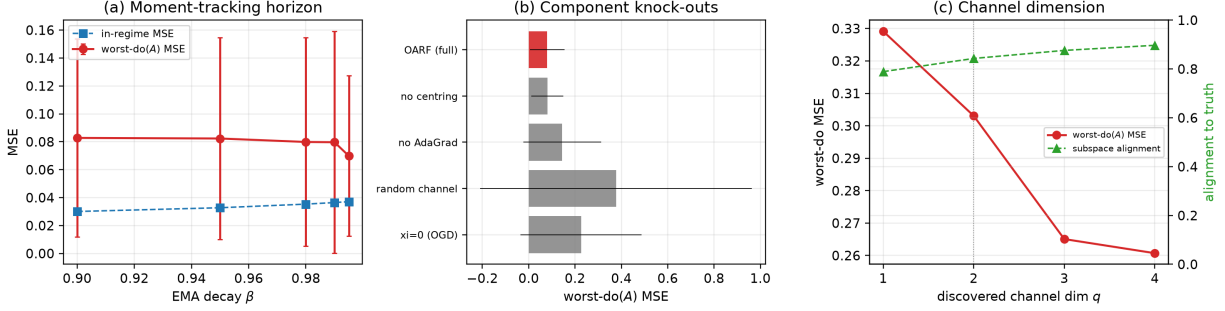


Figure 6: Ablations. (a) Sensitivity to the EMA decay β . (b) Component knock-outs by worst-case interventional MSE: the channel and the adaptive step are load-bearing, a random channel is worse than none. (c) Discovered-channel dimension: robustness and alignment both improve with q and are already strong at the true $q = 2$ (dotted).

Stress tests: when the anchor is wrong, OARF stops helping. The synthetic benchmark is, by construction, favourable: the anchor is clean and fully observed. The relevant question for practice is what happens when it is not, so we stress the fixed-channel method along two axes that break the ideal conditions (Figure 7; the synthetic SCM is otherwise unchanged, 8 seeds). *First*, we corrupt the observed context with additive noise of growing scale before any method sees it. Worst-do(A) MSE rises monotonically with the noise, and OARF crosses the no-anchor OGD floor at roughly one standard deviation of observation noise (0.18 vs. OGD 0.19): beyond that point a noisy anchor is no better than no anchor, and the learned channel degrades sooner. *Second*, we rotate the supplied channel away from the truth by an angle θ (subspace alignment $\cos \theta$). OARF beats OGD while the alignment exceeds about 0.6 (roughly $\theta \leq 50^\circ$) and is *worse* than OGD once the channel is more misaligned, reaching 0.36 versus OGD’s 0.19 at an orthogonal channel — the same mechanism as the random-channel row of Table 6. The practical message is a usable diagnostic rather than a reassurance: OARF should be deployed only when the analyst has reason to believe the anchor is informative (alignment well above 0.6, noise below order one); when the channel is weak, noisy, or guessed, anchoring is not a free hedge and can actively hurt, and a noise-only or unstable-discovery diagnostic — e.g. monitoring whether the learned channel stabilises — should gate its use.

Real data: competitive, but not significantly better. On the energy panel (Table 7) the honest conclusion is that OARF is *competitive but not a clear winner*. The central fact is statistical: on both targets *every* listed method lies within the 10% Model Confidence Set (Diebold–Mariano statistics in Figure 9 are mostly small), so the differences in Table 7 are *not* statistically significant and should not be read as establishing superiority. Within that caveat the point estimates are mildly favourable to the anchor methods on the regime-robustness metrics — on Brent, OARF-CD has the second-lowest worst-regime MSE (0.229) behind the batch/conformal methods and below OGD (0.233), DR-Combo (0.233), FC-DRO (0.235) and M-FISHER (0.269); on Henry Hub OARF-CD has the lowest worst-regime error among the online learners (0.385) — but on *overall* MSE ACI, Rolling-OLS and AnchorReg are at least as good, and ACI is the strongest single method on several metrics. We therefore do not claim OARF “leads” here; we claim it is never dominated and is on par with the best baselines, while M-FISHER and the Expected-Shortfall combiner are the weakest. The most likely explanation is that real energy variance has weaker, noisier driver–residual coupling than the controlled SCM — precisely the regime in which the stress tests above predict the anchor

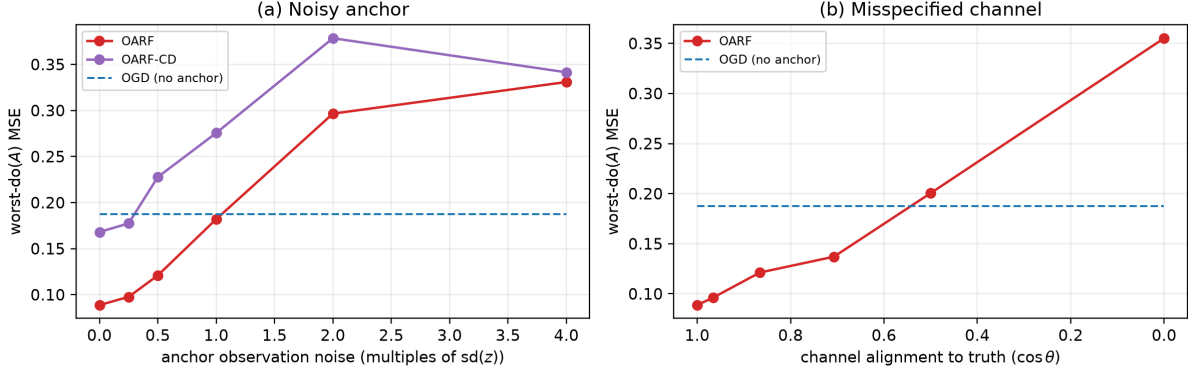


Figure 7: Stress tests of the fixed-channel method (worst-do(A) MSE; OGD floor dashed). (a) Increasing anchor observation noise: OARF crosses the no-anchor floor near one standard deviation of noise. (b) Rotating the channel away from the truth: anchoring helps only while alignment $\cos \theta \gtrsim 0.6$ and hurts beyond it. A wrong anchor is not a benign hedge.

advantage shrinks toward zero. The conformal head’s coverage is near nominal (0.90). A decision-economic vol-timing check is deferred to the appendix: over this sample the risk-targeting Sharpe ratios are low and statistically indistinguishable across methods, so it carries no evidential weight.

Table 7: Real energy log-realised-variance combination, 2010–2026 evaluation window (lower is better). Every listed method lies within the 10% Model Confidence Set; OARF and OARF-CD lead the online robust competitors on worst-regime MSE.

Method	Brent (DCOILBRENTU)				Henry Hub (DHHNGSP)			
	MSE	w-reg	post-cp	cov	MSE	w-reg	post-cp	cov
OARF	0.2225	0.2313	0.2384	—	0.2870	0.3928	0.2670	—
OARF-CD	0.2225	0.2292	0.2433	—	0.2857	0.3850	0.2627	—
OGD	0.2243	0.2334	0.2412	—	0.2860	0.3888	0.2655	—
Rolling-OLS	0.2203	0.2230	0.2358	—	0.2882	0.3916	0.2623	—
ACI	0.2180	0.2221	0.2399	0.900	0.2808	0.3833	0.2426	0.904
M-FISHER (2025)	0.2680	0.2691	0.3307	—	0.3532	0.4428	0.3442	—
DR-Combo (Wang 2026)	0.2304	0.2325	0.2507	—	0.2891	0.3958	0.2432	—
FC-DRO (Liu 2026)	0.2318	0.2346	0.2571	—	0.2872	0.3916	0.2436	—
FC-DRO-ES (Liu 2026)	0.2387	0.2440	0.2713	—	0.3056	0.4086	0.2689	—
W-DRO-OL (Chen 2026)	0.2243	0.2334	0.2406	—	0.2859	0.3890	0.2633	—
AnchorReg	0.2183	0.2195	0.2345	—	0.2907	0.3942	0.2716	—

Significance. The two Model Confidence Set analyses delineate what the evidence does and does not support. On the synthetic benchmark, the set computed on *in-distribution* loss (over 16 seeds) retains the batch and conformal methods — Rolling-OLS in all 16 seeds, DRIG in 15, the conformal heads in 11 — and excludes fixed-channel OARF (retained in 0 seeds) and OARF-CD (1 seed). This is the expected and correct outcome: OARF trades in-distribution accuracy for interventional robustness, which the in-distribution loss does not reward, so the in-distribution MCS is not the metric on which OARF should be judged — the held-out intervention grid is. On the real targets the MCS retains 10 of 11 methods on each commodity (only the weakest is excluded), so no method significantly out-predicts the field on average; we therefore treat the real-data differences

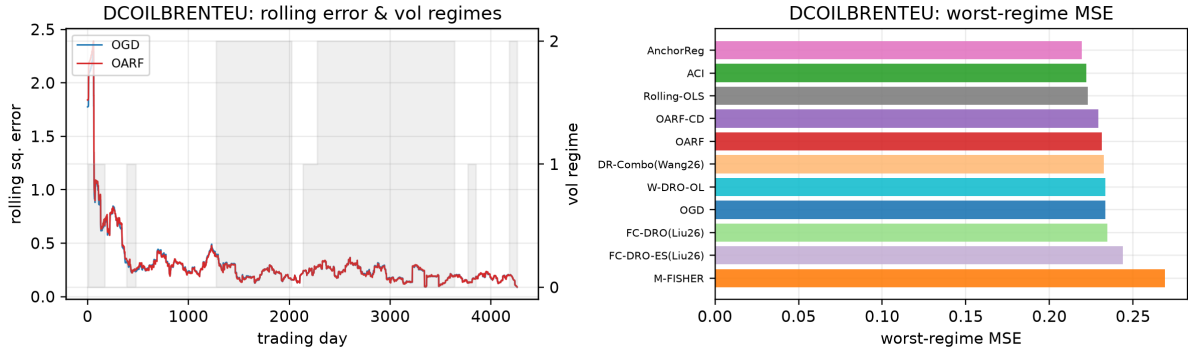


Figure 8: Brent: rolling squared error with the volatility-regime overlay (left) and worst-regime MSE by method (right).

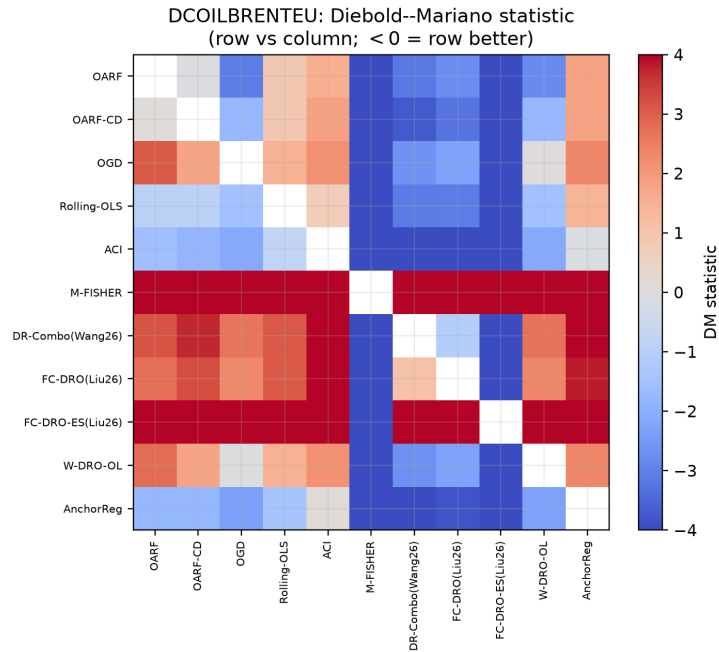


Figure 9: Brent: pairwise Diebold–Mariano statistics (negative = row more accurate than column). Most pairs are small in magnitude, consistent with the Model Confidence Set retaining all methods.

as descriptive, not significant. The Diebold–Mariano statistics corroborate this — pronounced on the synthetic benchmark (Figure 10) and mostly small on the real targets (Figure 9).

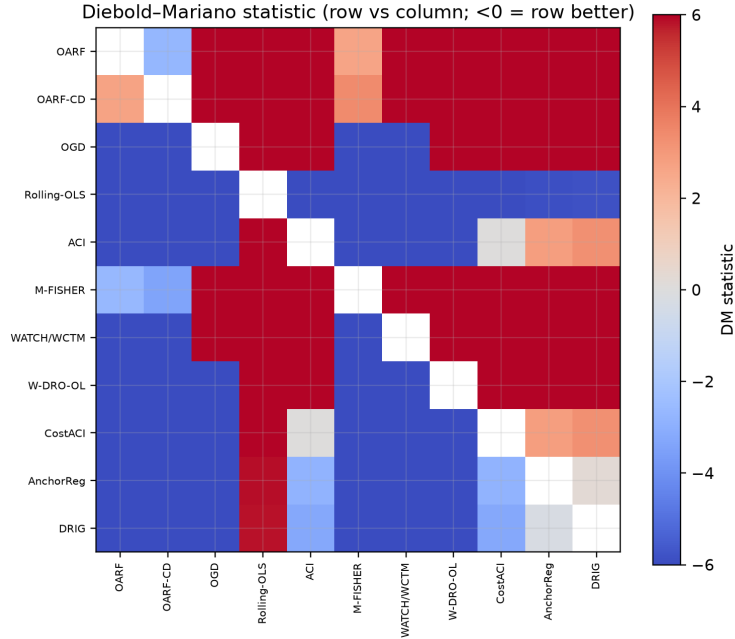


Figure 10: Pairwise Diebold–Mariano statistics on the synthetic benchmark; a negative entry means the row method is significantly more accurate than the column method.

6 Discussion and conclusion

The evidence supports a bounded claim. When regime shifts travel through an observable channel with anchor-mediated confounding, a forecaster can be immunised against them in advance and online, and the payoff is a large reduction in worst-case interventional error for a real but tunable increase in in-distribution error. When that structure is weak — as on the real energy series — the method is competitive but not significantly better than the strongest baselines. The geometric reading is that robustness is a budget: distributionally robust optimisation spends it isotropically, whereas anchoring concentrates it on the directions the regime actually moves; that concentration helps only to the extent that those directions are correctly identified.

When OARF should not be used. The same mechanism that helps under a good anchor hurts under a bad one, so it is essential to state the failure modes plainly. OARF is *not* recommended, and may underperform a plain online learner, when: (i) the supplied or discovered channel is misaligned with the true driver subspace (empirically, alignment below about 0.6; Figure 7b); (ii) the drivers are observed too noisily (observation noise above order one; Figure 7a); (iii) the dominant regime driver is *unobserved*, so no channel in z can capture it; (iv) the driver–residual coupling is weak, as it appears to be for daily energy variance, leaving little for anchoring to exploit; (v) the stream is short or regimes change faster than the EMA window can track, since both the moment estimates and the discovery layer need within-regime stationarity; and (vi) one relies on the learned channel for interventional robustness, where it is presently unstable. A practical safeguard is to monitor

channel stability and the anchor–residual coupling online and to fall back to $\xi = 0$ (plain OGD) when they indicate a weak or shifting channel.

Several limitations remain. The interventional-regret theorem is established for the *fixed*-channel learner; its EMA-moment rate of $T^{2/3}$ rests, through Lemma 3, on the within-regime stationarity of Assumption 4, and tightening it toward $\sqrt{T}$ without reintroducing a changepoint trigger, as well as extending the guarantee to the learned-channel variant whose channel drifts slowly with the data, is the principal open problem. Beyond it, a quantile or density head carrying *interventional* coverage guarantees, and a frictional-allocator head with decision-regret rather than forecast-regret bounds, would connect the method more tightly to the decisions that motivate it. We introduced OARF, a streaming anchor-robust forecaster that immunises against economic regime shifts through the causal channel they travel and that discovers that channel online; it is provably no-regret in interventional regret in the fixed-channel case ($O(\sqrt{T(1+K)})$ with oracle moments, $O((1+K)T^{2/3})$ online), cuts worst-case interventional error by more than half against the reactive and distributionally robust state of the art on a controlled benchmark, and holds its own over sixteen years of energy-volatility data while offering an interpretable, automatically discovered view of which drivers carry the regime.

Reproducibility. All code and data-build scripts that regenerate every number, table and figure in this paper are available at <https://github.com/vinhqddang/OARF-Online-Anchor-Robust-Forecasting-under-Economic-Regime-Shift>.

A Proof of Theorem 1

Throughout, $\mathcal{R}_t(w) := \sup_{\|v\| \leq \rho(\xi)} \mathbb{E}_{\text{do}(a:=v)}[(y_t - w^\top \phi(x_t))^2]$ is the per-step worst-case risk in (6), and $\hat{g}_t$ is the (inexact) step direction of Algorithm 1.

Lemma 1 (Convex surrogate and gradient decomposition). *$\mathcal{R}_t$ is convex, and under Assumption 2 it equals, within the active regime, the anchor-penalised risk $\mathcal{R}_t(w) = \mathbb{E}_t[(y - w^\top \phi)^2] + \xi P_t(w)$ of (1). Moreover $\hat{g}_t = \nabla \mathcal{R}_t(w_t) + \zeta_t$ with $\zeta_t = M_t + b_t$, where $M_t := -r_t \phi(x_t) - \mathbb{E}_t[-r\phi]$ is a martingale difference ($\mathbb{E}[M_t | \mathcal{F}_{t-1}] = 0$) and b_t is the deterministic ($\mathcal{F}_{t-1}$ -measurable) error of the EMA-estimated penalty gradient relative to $\nabla(\xi P_t)$.*

Proof. For fixed ν , $\mathbb{E}_{\text{do}(a:=\nu)}[(y - w^\top \phi)^2]$ is a convex quadratic in w ; a pointwise supremum of convex functions is convex. The identity is Proposition 1 within the stationary active regime. The prediction term $-r_t \phi(x_t)$ is an unbiased one-sample estimate of $\frac{1}{2} \nabla \mathbb{E}_t[(y - w^\top \phi)^2] = -\mathbb{E}_t[r\phi]$, with centred part M_t ; the robustness term substitutes the $\mathcal{F}_{t-1}$ -measurable EMA moments for the population moments in $\nabla(\xi P_t)$, and b_t is that substitution error. $\square$

Lemma 2 (Inexact dynamic regret). *Run projected OGD with the path-length-adaptive Ader wrapper [20] on the convex losses $\mathcal{R}_t$, fed $\hat{g}_t$. Then for any $u_{1:T} \subset \mathcal{W}$ of path length P_T , $\sum_t \mathcal{R}_t(w_t) - \sum_t \mathcal{R}_t(u_t) \leq O(\sqrt{T(1+P_T)}) + \sum_t \langle \zeta_t, w_t - u_t \rangle$, with the first term attained without prior knowledge of P_T .*

Proof. With exact gradients, Ader attains minimax dynamic regret $O(\sqrt{T(1+P_T)})$ for convex G -Lipschitz losses over a radius- D set, adaptively in P_T [20], subsuming Hazan [10], Zinkevich [21]. Convexity gives $\mathcal{R}_t(w_t) - \mathcal{R}_t(u_t) \leq \langle \nabla \mathcal{R}_t(w_t), w_t - u_t \rangle = \langle \hat{g}_t, w_t - u_t \rangle - \langle \zeta_t, w_t - u_t \rangle$; summing and applying the exact-gradient bound to the $\hat{g}_t$ terms yields the inexactness term. $\square$

Lemma 3 (Gradient-error control). *Under Assumptions 1–4, against the deterministic minimising comparator: (i) $\mathbb{E}[\sum_t \langle M_t, w_t - u_t \rangle] = 0$; and (ii) $\mathbb{E}\|b_t\| \leq c_1\sqrt{1-\beta}$ at steps at least $1/(1-\beta)$ into a regime, with $\|b_t\| \leq c_2$ within $1/(1-\beta)$ steps of each of the K changes, so $\sum_t \mathbb{E}\|b_t\| \leq c_1T\sqrt{1-\beta} + c_2K/(1-\beta)$; with oracle moments $b_t \equiv 0$.*

Proof. (i) w_t is $\mathcal{F}_{t-1}$ -measurable and, the schedule being oblivious (Assumption 3), u_t is deterministic, so $\mathbb{E}[\langle M_t, w_t - u_t \rangle \mid \mathcal{F}_{t-1}] = 0$; the tower rule concludes. (ii) An EMA of decay β of a stationary sequence has variance $\Theta(1-\beta)$ about its mean and geometrically-decaying within-regime bias; by Assumption 1 the penalty gradient is Lipschitz in these moments (the ϵ -ridge bounds the inverse), inheriting the $\Theta(\sqrt{1-\beta})$ steady-state scale. After a change the EMA mixes the previous regime for $O(1/(1-\beta))$ steps with bounded $\|b_t\| \leq c_2$. Summing gives the bound; oracle moments make the substitution error vanish. $\square$

Proof of Theorem 1. Combining the lemmas and taking expectations, for the deterministic minimiser $u_{1:T} \in \mathcal{W}$ ($P_T \leq 2DK$), $\mathbb{E}[\sum_t \mathcal{R}_t(w_t)] - \sum_t \mathcal{R}_t(u_t) \leq O(\sqrt{T(1+K)}) + 0 + D \sum_t \mathbb{E}\|b_t\|$. With oracle moments the last term vanishes, giving (i); with EMA moments it is $\leq D(c_1T\sqrt{1-\beta} + c_2K/(1-\beta))$, which at $1-\beta_T = T^{-2/3}$ is $O((1+K)T^{2/3})$ and dominates $\sqrt{T(1+K)}$, giving (ii). Finally, by Assumption 2 the realised regime is an admissible intervention, so $\mathbb{E}_t[\ell_t(w_t) \mid \mathcal{F}_{t-1}] \leq \mathcal{R}_t(w_t)$ and hence $\mathbb{E}[\sum_t \ell_t(w_t)] \leq \mathbb{E}[\sum_t \mathcal{R}_t(w_t)]$; as $\min_{u \in \mathcal{W}} \sum_t \mathcal{R}_t(u)$ is the comparator in (6), the displays bound $\mathbb{E}[\text{IntReg}_T(\rho(\xi))]$ by the stated rates. $\square$

B Decision-economic evaluation (secondary)

As a secondary check we convert each method’s volatility forecast into a volatility-timing portfolio [7]: exposure is scaled inversely to the forecast variance, $w_t = \text{clip}(\sigma^{*2}/\hat{\sigma}_t^2, 0, 3)$, and the strategy earns $w_t r_{t+1}$ net of a one-basis-point turnover cost; we report the annualised Sharpe ratio, maximum drawdown and turnover. Over 2010–2026 these commodity risk-targeting Sharpe ratios are low and statistically indistinguishable across methods (Figure 11), so this evaluation carries no evidential weight and is reported only for completeness; it is not used to support any claim in the main text.

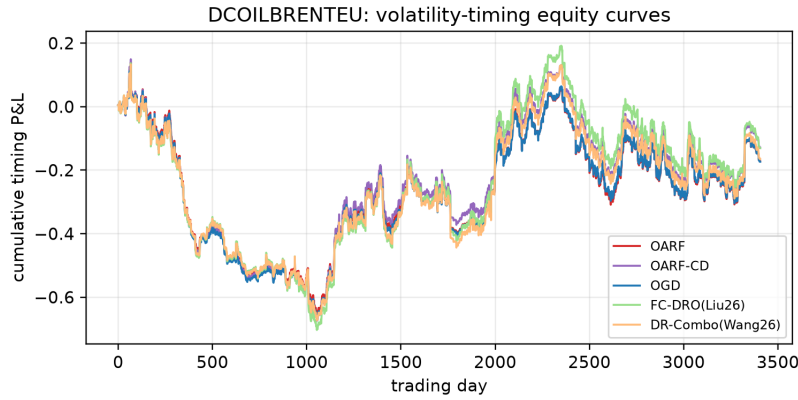


Figure 11: Brent: cumulative volatility-timing profit and loss. Differences across methods are economically small and not significant over this sample.

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